Computers, Ethics, and Social Values

Dr. Sherif M. Tawfik

Learning Objectives

1. **Introduction**
2. **Ethics in Society**
3. **Definition of computer ethics**
4. **Rationale for studying computer ethics**
5. **History of Computer Ethics**
6. **The Ten Commandments of computer ethics**
7. **Computer Ethics Topics**
   - ❖ **Computers in the Workplace**
   - ❖ **Privacy and Anonymity**
   - ❖ **Computer Crime**
   - ❖ **Intellectual Property**
   - ❖ **Professional Responsibility**
   - ❖ **Globalization**
   - ❖ **The computer security**

Learning Objectives

8. **Case Studies**
9. **Why are we discussing this?**
10. **Organizations of Computing Professionals That Have Ethical Codes**
11. **Common Themes of Computer Society Codes**
12. **ACM (Code of Ethics and Professional Conduct)**

Introduction

- Most everyone shares "core values", desiring:
  - Life
  - Happiness
  - Ability to accomplish goals
- Two ways to view world
  - **Selfish point of view:** consider only your own self and your core values
  - **Ethical point of view:** respect other people and their core values

Ethics in Society

- **Society**
  - An association of people organized under a system of rules designed to advance the good of its members over time
  - **Ethics vs. Morals**. ... While they are sometimes used **interchangeably,** they are **different**: **ethics** refer to rules provided by an external source, e.g., codes of conduct in workplaces or principles in religion religions. **Morals** refer to an individual's own principles regarding right and wrong.

Computer Ethics

- Computer ethics deals with the procedures, values and practices that govern the process of consuming computing technology and its related disciplines without damaging or violating the moral values and beliefs of any individual, organization or entity.
- In another words, Computer ethics is a concept in ethics that addresses the ethical issues and constraints that arise from the use of computers, and how they can be mitigated or prevented.

Rationale for studying Computer Ethics

- New entities: programmes, software, microchips, Websites, video games, the Cloud
- Scale of organised activity: data collection, calculation, statistical analysis
- Scale of calculations: new types of knowledge in fields such as meteorology, economics, military technology
- Inherent unreliability of computer systems: new ways of thinking about risk, accountability and liability
- Power and pervasiveness: dependency on computers for all aspects of modern living.

Rationale for studying Computer Ethics

- As in the previous slide, Computer and information technology raises many (often new) ethical questions about the right ways of uses of this new technology.
- Most people, including computer professionals, are vaguely aware of these technology related ethical questions.
- Policies and laws lag behind the technology.
- Extending the ordinary or traditional moral principles to situations created by computer and information technology is often awkward and/or inadequate.

History of Computer Ethics

- MIT professor Norbert Wiener WWII (early 1940s)

“second industrial revolution “

the integration of computer technology into society will eventually constitute the remaking of society

History of Computer Ethics

Mid 1960s, Donn Parker of SRI

"when people entered the computer center they left their ethics at the door"

Late 1960s, Joseph Weizenbaum (MIT)

inspired many thinkers and projects in computer ethics

Mid 1970s, Walter Maner

inquiry dealing with ethical problems aggravated, transformed or created by computer technology

History of Computer Ethics

By the 1980s, a number of social and ethical consequences of IT going public:

- **computer-enabled crime**
- **disasters caused by computer failures**
- **invasions of privacy via computer databases**
- **major law suits regarding software ownership**

History of Computer Ethics

Mid 80's

James Moor (Dartmouth College) published **influential article "*What Is Computer Ethics?*"**

Deborah Johnson published *Computer Ethics*, the first textbook

History of Computer Ethics

1990's

new university courses, research centers, conferences, journals, articles and textbooks appeared, and a wide diversity of additional scholars and topics became involved.

Computer Professionals for Social Responsibility

The Ten Commandments of computer ethics

1. Don’t use a computer to harm other people.
2. Don’t interfere with other people's computer work.
3. Don’t snoop around in other people's computer files.
4. Don’t use a computer to steal.
5. Do not contribute to the spread of misinformation using computer technology.
6. Don’t copy or use proprietary software for which you have not paid (without permission).

The Ten Commandments of computer ethics

7. Don’t use other people's computer resources without authorization or proper compensation.
8. Don’t appropriate other people's intellectual output.
9. Think about the social consequences of the program you are writing or the system you are designing.
10. Always use a computer in ways that ensure consideration and respect for your fellow humans

The Ten Commandments of computer ethics

1. Don’t use a computer to harm other people.
   - This commandment says that it is unethical to use a computer to harm another user. It is not limited to physical injury. It includes harming or corrupting other users' data or files.
   - The commandment states that it is wrong to use a computer to steal someone's personal information. Manipulating or destroying files of other users is ethically wrong. It is unethical to write programs, which on execution lead to stealing, copying or gaining unauthorized access to other users' data.
   - Being involved in practices like hacking, spamming, phishing or cyber bullying does not conform to computer ethics.

The Ten Commandments of computer ethics

2. Don’t interfere with other people's computer work.

- Computer software can be used in ways that disturb other users or disrupt their work.
- Viruses, for example, are programs meant to harm useful computer programs or interfere with the normal functioning of a computer.
- Malicious software can disrupt the functioning of computers in more ways than one. It may overload computer memory through excessive consumption of computer resources, thus slowing its functioning. It may cause a computer to function wrongly or even stop working.
- Using malicious software to attack a computer is unethical.

The Ten Commandments of computer ethics

3. Don’t snoop around in other people's computer files.
   - We know it is wrong to read someone's personal letters. On the same lines, it is wrong to read someone else's email messages or files.
   - Obtaining data from another person's private files or reading someone else's personal messages is the invasion of his privacy.
   - There are exceptions to this. For example, spying is necessary and cannot be called unethical when it is done against illegitimate use of computers. For example, intelligence agencies working on cybercrime cases need to spy on the internet activity of suspects.

The Ten Commandments of computer ethics

4. Don't use a computer to steal.
   - Stealing sensitive information or leaking confidential information is as good as robbery. It is wrong to acquire personal information of employees from an employee database or patient history from a hospital database or other such information that is meant to be confidential.
   - Similarly, breaking into a bank account to collect information about the account or account holder is wrong.
   - Illegal electronic transfer of funds is a type of fraud. With the use of technology, stealing of information is much easier. Computers can be used to store stolen information.

The Ten Commandments of computer ethics

5. Do not contribute to the spread of misinformation using computer technology.
   - Spread of information has become viral today, because of the Internet. This also means that false news or rumors can spread speedily through social networking sites or emails.
   - Being involved in the circulation of incorrect information is unethical. Mails and pop-ups are commonly used to spread the wrong information or give false alerts with the only intent of selling products.
   - Mails from untrusted sources advertising certain products or spreading some hard-to-believe information, are not uncommon. Direct or indirect involvement in the circulation of false information is ethically wrong. Giving wrong information can hurt other parties or organizations that are affected by that particular theme.

The Ten Commandments of computer ethics

6. Don't copy or use proprietary software for which you have not paid (without permission).
   - Like any other artistic or literary work, software is copyrighted.
   - A piece of code is the original work of the individual who created it. It is copyrighted in his/her name.
   - In case of a developer writing software for the organization she works for, the organization holds the copyright for it.
   - Copyright holds true unless its creators announce it is not. Obtaining illegal copies of copyrighted software is unethical and also encourages others to make copies illegally.

The Ten Commandments of computer ethics

7. Don’t use other people's computer resources without authorization or proper compensation.
   - Multi-user systems have user specific passwords.
   - Breaking into some other user's password, thus intruding his/her private space is unethical.
   - It is not ethical to hack passwords for gaining unauthorized access to a password-protected computer system.
   - Accessing data that you are not authorized to access or gaining access to another user's computer without her permission is not ethical.

The Ten Commandments of computer ethics

8. Don’t appropriate other people's intellectual output.
   - Programs developed by a software developer are her property.
   - If he is working with an organization, they are the organization's property.
   - Copying them and propagating them in one's own name is unethical.
   - This applies to any creative work, program or design. Establishing ownership on a work which is not yours is ethically wrong.

The Ten Commandments of computer ethics

9. Think about the social consequences of the program you are writing or the system you are designing.
   - A computer software on release, reaches millions. Software like video games and animations or educational software can have a social impact on their users.
   - When working on animation films or designing video games, for example, it is the programmer's responsibility to understand his target audience/users and the effect it may have on them.
   - For example, a computer game for kids should not have content that can influence them negatively. Similarly, writing malicious software is ethically wrong.
   - A software developer/development firm should consider the influence their code can have on the society at large.

The Ten Commandments of computer ethics

10. Always use a computer in ways that ensure consideration and respect for your fellow humans

- The communication etiquette we follow in the real world applies to communication over computers as well.
- While communicating over the Internet, one should treat others with respect.
- One should not intrude others' private space, use abusive language, make false statements or pass irresponsible remarks about others.
- One should be courteous while communicating over the web and should respect others' time and resources.
- Also, one should be considerate with a novice computer user.

Computer Ethics Topics

- Computers in the Workplace
- Privacy and Anonymity
- Computer Crime
- Intellectual Property
- Professional Responsibility
- Globalization
- The computer security

Computers in the Workplace

- LOST JOBS: Economic incentives to replace humans with computerized devices are very high
- NEW JOBS: Computer industry already has generated a wide variety of new jobs
- ALTERED JOBS:
  - "de-skilling"
  - New skilled jobs

Privacy and Anonymity

- Mid-1960s

  the American government already had created large databases of information about private citizens - census data, tax records, military service records, welfare records, and so on.

This raise the importance of information systems ethics

Information Systems Ethics

- Privacy
  - Protecting one's personal information
- Identity theft
  - Stealing of another's social security number, credit card number, or other personal information
- Information accuracy
  - Deals with authentication and fidelity of information
- Information property
  - Deals with who owns information about individuals and how information can be sold and exchanged

Information Systems Ethics

- Information accessibility
  - Deals with what information a person has the right to obtain about others and how the information can be used
- Issues in information accessibility
  - Carnivore: software application designed to be connected to Internet Service Providers' computers and eavesdrops on all communications.
  - Electronic Communications Privacy Act (ECPA): it offered stronger support for voice mail than it did for e-mail. No other laws at federal or state levels protect e-mail privacy
  - Monitoring e-mail

Information Systems Ethics

- The need for a code of ethical conduct
  - Business ethics
  - Plagiarism
  - Cybersquatting: registering a domain name and then trying to sell the name for big bucks to a person, company. Domain names are a scarce resource – one of the few scarce resources in cyberspace

Computer Crime

- **Definition: the act of using a computer to commit an illegal act**
  - Authorized and unauthorized computer access
  - Examples
    - Stealing time on company computers
    - Breaking into government Web sites
    - Stealing credit card information

Computer Crime

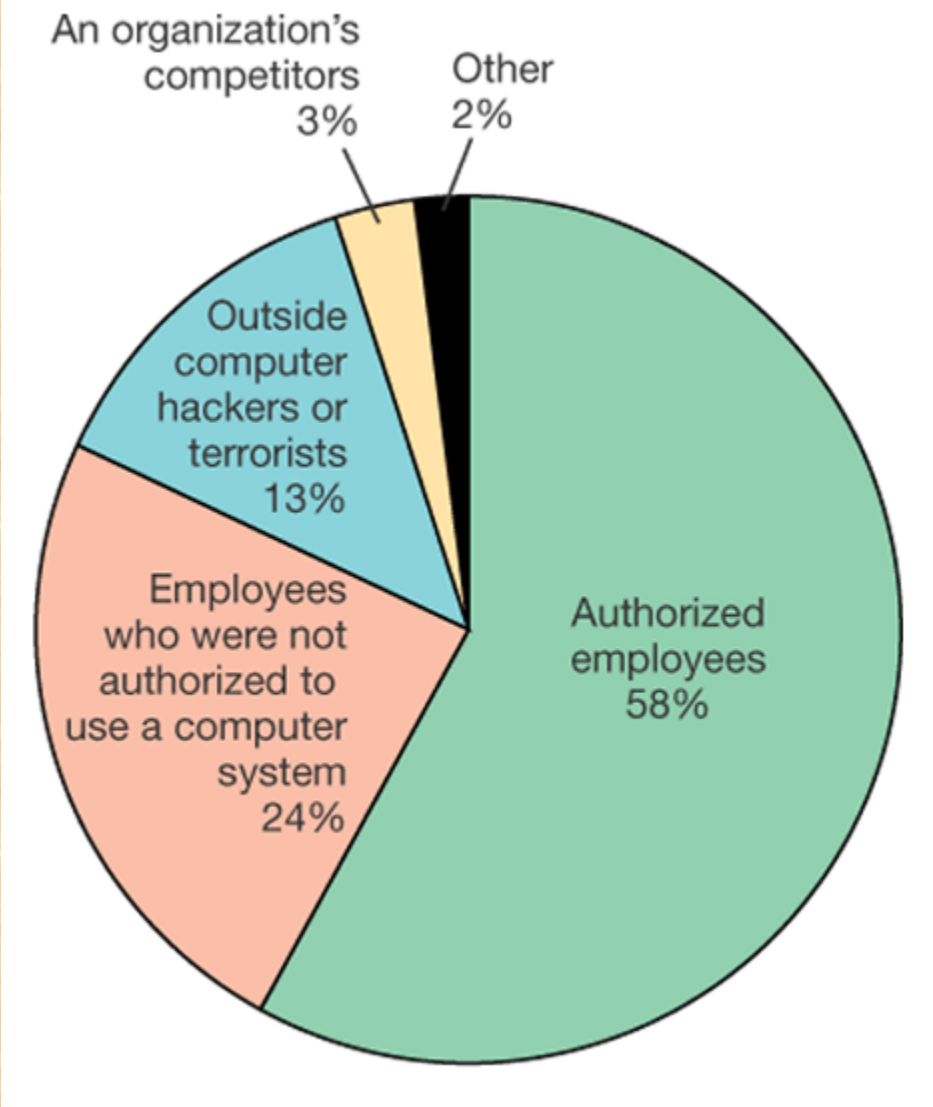
- **Federal and State Laws**
  - Stealing or compromising data
  - Gaining unauthorized computer access
  - Violating data belonging to banks
  - Intercepting communications
  - Threatening to damage computer systems
  - Disseminating viruses

Computer Crime

- **Hacking and Cracking**
  - Hacker – one who gains unauthorized computer access, but without doing damage
  - Cracker – one who breaks into computer systems for the purpose of doing damage

Computer Crime

- **Who commits computer crime?**



Computer Crime

- **Computer viruses and destructive code**
  - Virus – a destructive program that disrupts the normal functioning of computer systems
  - Types:
    - Worm: usually does not destroy files; copies itself
    - Trojan horses: Activates without being detected; does not copy itself
    - Logic or time bombs: A type of Trojan horse that stays dormant for a period of time before activating

Intellectual Property

- Free Software Foundation - software ownership should not be allowed at all.
- One can own the following aspects of a program:
  - source code
  - object code
  - algorithm

Professional Responsibility

- Computer professionals find themselves in a variety of professional relationships with other people
- Association for Computing Machinery (ACM) and the Institute of Electrical and Electronic Engineers (IEEE), have established codes of ethics, curriculum guidelines and accreditation requirements

Computer Security

- Computer Security – precautions taken to keep computers and the information they contain safe from unauthorized access

Case Study # 1

- As a computer programmer working at a bank, you discover that you have accidentally been given write access to payroll data for all bank employees. Do you give yourself a raise?

Case Study # 2

- As a computer programmer working at a bank, you discover that you have accidentally been given read access to payroll data for all bank employees. Do you compare your pay to that of other programmers?

Case Study # 3

- As a data base administrator, you have been given read access to personnel data. Do you search for the private home phone number of the person you would most like to get to know?

Case Study # 4

- Your nerdy roommate has carelessly left his password where you can find it. Do you threaten Hillary Clinton (or George Bush) using his account?

Case Study # 5

- Your best friend offers to let you copy the latest version of Windows. Do you?

Case Study # 6

- Consider this statement of Mr. Jonah Nonimous: “I’m a law-abiding citizen, I pay my taxes promptly. I have a loving family. I don’t care if anyone reviews my college grades or income tax records, because I have nothing to hide. All these privacy laws are unnecessary. Only individuals who have something to hide need them.”

Case Study # 7

- The past several months, George, an electrical engineer working for an aerospace contractor, has been the quality control manager on a project to develop a computerized control system for a new military aircraft. Early simulations of the software for the control system showed that, under certain conditions, instabilities would arise that would cause the plane to crash. The software was subsequently patched to eliminate the specific problems uncovered by the tests. After the repairs were made, the system passed all of the required simulation tests.
- George is convinced, however, that those problems were symptomatic of a fundamental design flaw that could only be eliminated by an extensive redesign of the system. Yet, when he brought his concern to his superiors, they assured him that the problems had been resolved, as shown by the tests. Anyway, to reevaluate and possibly redesign the system would introduce delays that would cause the company to miss the delivery date specified in the contract, and that would be very costly.

Case Study # 7

- Now, there's a great deal of pressure on George to sign off on the system and allow it to be flight tested. It has even been hinted that, if he persists in delaying release of the system, the responsibility will be taken away from him and given to someone who is more compliant. . . .

- What makes the situation so difficult for George is that he must choose between conflicting duties: loyalty to self, family, employer, and superiors versus the obligation to tell the truth and to protect others from harm. . .

Why are we discussing this?

- Because you are working toward becoming a computer professional who will have knowledge that many do not, who will have access to information that many do not, who will have the ability to cause harm or to do good.

Why are we discussing this?

- Because as a teacher of a powerful, potentially harmful set of knowledge, I have an obligation to impart a sense of responsibility to my students.
- Because our accrediting agency encourages us to do so.

Organizations of Computing Professionals That Have Ethical Codes

- Association of Information Technology Professionals (AITP)
- Institute for the Certification of Computer Professionals (ICCP)
- Association for Computing Machinery (ACM)
- Institute for Electrical and Electronics Engineers (IEEE) Computer Society
- British Computer Society (BCS)
- Canadian Information Processing Society (CIPS)
- Independent Computer Consultants Association (ICCA)

Links to Organizations of Computing Professionals That Have Ethical Codes

- AITP http://www.aitp.org/index.jsp
- ICCP http://www.iccp.org/iccpnew/ethics%20practice%20conduct.html
- ACM http://www.acm.org/constitution/code.html
- IEEE http://www.ieee.org/about/whatis/code.html
- BCS http://www.bcs.org/
- CIPS http://www.cips.ca/about/ethics/
- ICCA http://www.icca.org/ethics.asp

Common Themes of Computer Society Codes

- Dignity and worth of other people
- Personal integrity and honesty
- Responsibility for work
- Confidentiality of information
- Public safety, health, and welfare
- Participation in professional societies to improve the profession
- Knowledge and access to technology = social power

ACM Code of Ethics and Professional Conduct

- **General moral imperatives**
  - contribute to society and human well-being
  - avoid hard to others
  - be honest and trustworthy;
  - be fair and take action not to discriminate
  - honor property rights including copyright and patent
  - give proper credit for intellectual property
  - respect the privacy of others
  - honor confidentiality

ACM Code of Ethics and Professional Conduct

- **More specific professional responsibilities**
  - Strive to achieve highest quality, effectiveness and dignity in both the process and products of professional work
  - Acquire and maintain professional competence
  - Know and respect existing laws pertaining to professional work
  - Accept and provide appropriate professional review
  - Give comprehensive and thorough evaluations of computer systems and their impacts, including analysis of possible risks.
  - Honor contracts, agreements, and assigned responsibilities
  - Improve public understanding of computing and its consequences
  - Access computing and communication resources only when authorized to do do

ACM Code of Ethics and Professional Conduct

- **Organizational leadership imperatives**
  - Articulate social responsibilities of members of an organizational unit and encourage full acceptance of those responsibilities
  - Manage personnel and resources to design and build information systems that enhance the quality of working life.
  - Acknowledge and support proper and authorized uses of an organization's computing and communication resource.
  - Ensure that users and those who will be affected by a system have their needs clearly articulated during the assessment and design of requirements; later the system must be validated to meet requirement.
  - Articulate and support policies that protect the dignity of users and others affected by a computing system.
  - Create opportunities for members of the organization to learn the principles and limitations of computer systems.